$$\hat{R}_n = \frac{1}{n}\sum_n \hat{l}(g(\mathbf{w},\mathbf{x}),y)$$

$$\hat{R}_n(h) = \frac{1}{n}\sum_{i=0}^{n}(h(\mathbf{w},\mathbf{x}_i)-y_i)^2$$
$$\hat{R}_n(\mathbf{w}) = \frac{1}{n}\sum_{i=0}^{n}(\mathbf{w}^T\mathbf{x}_i-y_i)^2$$
$$\hat{R}_n(\mathbf{w}) = \frac{1}{n}\|X\mathbf{w}-\mathbf{y}\|^2$$

$$\hat{R}_{classification}(\mathbf{w}) = \frac{1}{n}\sum_i \mathbb{I}[\text{sign}(\mathbf{w}^T\mathbf{x}_i)\neq y_i]$$

$$g = \underset{h\in\mathcal{H}}{\text{argmin}}\left\{R(h)=\mathbb{E}\left[\ell(h(\mathbf{x}),y)\right]\right\}$$

$$|\hat{R}_n(h)-R(h)| \leqslant \sqrt{\frac{\log\frac{2M}{\delta}}{2n}}$$

$$\hat{R}_n(h) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{I}\left(h(x_i)\neq f(x_i)\right)$$

$$\mathbf{x}=(x_0,x_1,\ldots,x_d)\xrightarrow{\Phi}\mathbf{z}=(z_0,z_1,\ldots,z_{\tilde{d}})$$

$$\mathbf{w}_{lin} = (Z^\top Z)^{-1}Z^\top\mathbf{y}$$

$$\mathcal{L}_n(w) = \hat{R}_n(\mathbf{w}) + \lambda\|\mathbf{w}\|_Q^2$$

$$\theta(s) = \frac{e^s}{1+e^s} = \frac{1}{1+e^{-s}}$$

$$\Delta\mathbf{w} = -\eta\nabla\hat{R}_n(\mathbf{w}_t) \text{ (with } \eta \text{ redefined)}$$

$$\delta_1^{(L)} = \frac{\partial\ell_k}{\partial s_1^{(L)}} = \frac{\partial\ell_k}{\partial x_1^{(L)}}\cdot\frac{\partial x_1^{(L)}}{\partial s_1^{(L)}}$$

$$m_{\mathcal{H}} \leqslant \sum_{i=0}^{d_{VC}}\binom{n}{i}$$

$$\frac{d}{dx}\tanh(x) = 1-\tanh^2(x).$$

$\sum_{i=0}^{d} w_i x_i \geqslant 0 \rightarrow \text{sign}(\sum_{i=0}^{d} w_i x_i)$, with $x_0 = 1$

Predictor: $h(\mathbf{x},\mathbf{w}) = \text{sign}(\mathbf{w}^T\mathbf{x})$

$$\mathbf{w} = (X^TX)^{-1}X^T\mathbf{y} = \dot{X}\mathbf{y}$$

$$P\left(|R(h)-\hat{R}_n(h)|>\varepsilon\right) \leqslant 2e^{-2\varepsilon^2 n}$$

$$P\left(|\hat{R}_n(h)-R(h)|>\varepsilon\right) \leqslant 2Me^{-2\varepsilon^2 n}$$

$$g = \underset{h\in\mathcal{H}}{\text{argmin}}\,\hat{R}_n(h)$$

$$R(h) = P\left(h(x_i)\neq f(x_i)\right)$$

$$\Phi(x) = \sum_{k=0}^{q} a_k x^k$$

$$\mathcal{L}_n(\mathbf{w}) = \hat{R}_n(\mathbf{w}) + \tfrac{\lambda}{n}\|\mathbf{w}\|_2^2 \quad (\lambda>0)$$

$$\log\frac{P(y=+1|x_1,\ldots,x_d)}{P(y=-1|x_1,\ldots,x_d)} = \mathbf{w}_*^\top\mathbf{x} = h(\mathbf{x},\mathbf{w})$$

$$P(y|\mathbf{x}) = \begin{cases}\theta(\mathbf{w}_*^\top\mathbf{x}) & \text{if } y=+1;\\ 1-\theta(\mathbf{w}_*^\top\mathbf{x}) & \text{if } y=-1,\end{cases}$$

$$\theta(s) = \max\{0,s\}$$

$$H(w,b) = \{x : w^\top x + b = 0\}$$

$$m_{\mathcal{H}}(n) = 2^n$$

While there exists a missclassified data point with $\text{sign}(\mathbf{w}^\top\mathbf{x}_i)\neq y_i \qquad y\in\{-1,+1\}$

update $\mathbf{w}\leftarrow\mathbf{w}+y_i\mathbf{x}_i$

$$\hat{R}_{regression}(\mathbf{w}) = \frac{1}{n}\sum_i(\mathbf{w}^T\mathbf{x}_i-y_i)^2$$

$$P\left(\sup_{h\in\mathcal{H}}|\hat{R}_n(h)-R(h)|\right)>\varepsilon) \leqslant 4\underbrace{m_{\mathcal{H}}(2n)}_{\leqslant(2n+1)^{d_{VC}(\mathcal{H})}}e^{-\varepsilon^2 n/8}$$

$$\delta = 2Me^{-2\varepsilon^2 n} \quad\Rightarrow\quad \varepsilon = \sqrt{\frac{\log\frac{2M}{\delta}}{2n}}$$

$$R(g)-R(h^*) \leqslant \sqrt{\frac{2\log\frac{2M}{\delta}}{n}}$$

$$R(h) \leqslant \hat{R}_n(h) + \underbrace{\sqrt{\frac{8d_{VC}(\mathcal{H})}{n}\log(2n+1)+\frac{8}{n}\log\frac{4}{\delta}}}_{\Omega(n,\mathcal{H},\delta)}$$

$$\hat{R}_n = \frac{1}{n}\sum_{i=1}^{n}(w_i z_i - y_i)^2 = \frac{1}{n}(Z\mathbf{w}-\mathbf{y})^\top(Z\mathbf{w}-\mathbf{y})$$

$$\mathbf{w}_{reg} = (Z^\top Z+\lambda I)^{-1}Z^\top\mathbf{y}$$

$$P(y=1|\mathbf{x}) = \frac{e^{\mathbf{w}_*^\top\mathbf{x}}}{1+e^{\mathbf{w}_*^\top\mathbf{x}}} = \frac{1}{1+e^{-\mathbf{w}_*^\top\mathbf{x}}} = \theta(\mathbf{w}_*^\top\mathbf{x})$$

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \Delta\mathbf{w}$$

$$s_j^{(l-1)} = \sum_{i=0}^{d^{(l-2)}} w_{ij}^{(l-1)}x_i^{(l-2)}$$

Distance of x from H is $\frac{|w^\top x+b|}{\|w\|}$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \frac{d}{dx}\sigma(x) = \sigma(x)(1-\sigma(x))$$

$$\delta^* \triangleq \min_i \frac{|w^\top x + b|}{\|w\|} = \min_i \frac{y_i(w^\top x_i + b)}{\|w\|}$$

$$y^*(w^T x^* + b) = 1$$

$$(w_*, b_*) = \text{argmax}_{w,b:} \left\{ \frac{1}{\|w\|} \text{ such that } y_i(w^T x_i + b) \geq 1, \ \forall i \right\}$$

Once α_is are solved:

$$w^* = \sum_{i:\alpha_i^* > 0} \alpha_i^* y_i x_i$$

$$b^* = \frac{1}{y_i} - (w^*)^\top x_i$$

$$\max_\alpha \mathcal{L}(\alpha) \triangleq \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i y_j \alpha_i \alpha_j x_i^\top x_j$$

subject to $\alpha_i \geq 0$, $\sum_{i=1}^n \alpha_i y_i = 0$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 + \sum_{i=1}^n \alpha_i [1 - y_i(w^\top x_i + b)]$$

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2}\|w\|^2 + C\sum_{i=1}^n \xi_i + \sum_{i=1}^n \alpha_i [1 - \xi_i - y_i(w^\top x_i + b)] - \sum_{i=1}^n \beta_i \xi_i$$

$$\mathcal{L}(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n y_i y_j \alpha_i \alpha_j K(x_i, x_j)$$

$$\Phi(x)_n = \frac{1}{\sqrt{n!}} \exp -x^2/2x^n$$

$$v_c = \sum_{i \in knn(x)} \mathbb{I}(y_i = c), \quad \forall c \in \mathcal{C}$$

$$nn(x) = \underset{i \in [m]}{\text{argmin}} \|x - x_i\|^2$$

$$y = f(x) = \underset{c \in \mathcal{C}}{\text{argmax}} \, v_c$$

$$\|x - x'\|_p = \left(\sum_{j=1}^d |x_j - x'_j|^p \right)^{1/p}$$

$$\boldsymbol{x_i} = \sum_{j=1}^k w_{i,j} \boldsymbol{c_j}$$

$$\boldsymbol{c_i}^T \boldsymbol{c_j} = \begin{cases} 1 & \text{if } i = j; \\ 0 & \text{if } i \neq j. \end{cases}$$

$$\min_{C, \boldsymbol{w_1}, \ldots, \boldsymbol{w_n}} \frac{1}{n} \sum_{i=1}^n \|C\boldsymbol{w_i} - \boldsymbol{x_i}\|^2$$

$$\min_{C: C^T C = I_{d \times d}} \frac{1}{n} \sum_{i=1}^n \|CC^T \boldsymbol{x_i} - \boldsymbol{x_i}\|^2$$

$$\min_C \frac{1}{n} \sum_{i=1}^n \boldsymbol{q_i}^T (D - I_{d \times d})^2 \boldsymbol{q_i}$$

$$\frac{1}{n} \sum_{i=1}^n \left(\boldsymbol{c}^T \boldsymbol{x_i} - \boldsymbol{c}^T \bar{\boldsymbol{x}} \right)^2$$

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \left(\boldsymbol{c}^T \boldsymbol{x_i} \right)^2 &= \frac{1}{n} \sum_{i=1}^n \boldsymbol{c}^T (\boldsymbol{x_i} \boldsymbol{x_i}^T) \boldsymbol{c} \\ &= \boldsymbol{c}^T \left(\frac{1}{n} \sum_{i=1}^n \boldsymbol{x_i} \boldsymbol{x_i}^T \right) \boldsymbol{c} \\ &= \boldsymbol{c}^T \left(\frac{1}{n} XX^T \right) \boldsymbol{c} \end{aligned}$$

$$\begin{aligned} \|CC^T \boldsymbol{x_i} - \boldsymbol{x_i}\|^2 &= \boldsymbol{x_i}^T CC^T CC^T \boldsymbol{x_i} - 2\boldsymbol{x_i}^T CC^T \boldsymbol{x_i} + \boldsymbol{x_i}^T \boldsymbol{x_i} \\ &= -\boldsymbol{x_i}^T CC^T \boldsymbol{x_i} + \boldsymbol{x_i}^T \boldsymbol{x_i} \\ &= -\|C^T \boldsymbol{x_i}\|^2 + \|\boldsymbol{x_i}\|^2 \end{aligned}$$

$$\min_{\substack{\boldsymbol{c_1}, \ldots, \boldsymbol{c_k}: \\ \|\boldsymbol{c_i}\|^2 = 1, \\ \boldsymbol{c_i}^T \boldsymbol{c_j} = 0, i \neq j}} -\frac{1}{n} \sum_{i=1}^n \sum_{j=1}^k \left(\boldsymbol{c_j}^T \boldsymbol{x_i} \right)^2$$

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (\boldsymbol{c_1}^T \boldsymbol{x_i})^2 &= \frac{1}{n} \sum_{i=1}^n \boldsymbol{c_1}^T \boldsymbol{x_i} \boldsymbol{x_i}^T \boldsymbol{c_1} \\ &= \boldsymbol{c_1}^T \left(\frac{1}{n} \sum_{i=1}^n \boldsymbol{x_i} \boldsymbol{x_i}^T \right) \boldsymbol{c_1} \\ &= \boldsymbol{c_1}^T \left(\frac{1}{n} XX^T \right) \boldsymbol{c_1} \end{aligned}$$

$$\mathcal{L}(\boldsymbol{c_1}, \alpha) = -\boldsymbol{c_1}^T \left(\frac{1}{n} XX^T \right) \boldsymbol{c_1} + \alpha(\boldsymbol{c_1}^T \boldsymbol{c_1} - 1)$$

$$\boldsymbol{c_1}^T \left(\frac{1}{n} XX^T \right) \boldsymbol{c_1} = \boldsymbol{c_1}^T \alpha \boldsymbol{c_1} = \alpha$$

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \boldsymbol{y_i} \boldsymbol{y_i}^T &= \frac{1}{n} \sum_{i=1}^n L^{-1/2} U^T (\boldsymbol{x_i} - \bar{\boldsymbol{x}})(\boldsymbol{x_i} - \bar{\boldsymbol{x}})^T U L^{-1/2} \\ &= L^{-1/2} U^T \left(\frac{1}{n} \sum_{i=1}^n (\boldsymbol{x_i} - \bar{\boldsymbol{x}})(\boldsymbol{x_i} - \bar{\boldsymbol{x}})^T \right) U L^{-1/2} \\ &= L^{-1/2} U^T S U L^{-1/2} \\ &= L^{-1/2} U^T U L U^T U L^{-1/2} \\ &= L^{-1/2} L L^{-1/2} = I_{d \times d} \end{aligned}$$

$$\mathcal{L}(\boldsymbol{c_2}, \theta, \beta) = -\boldsymbol{c_2}^T \left(\frac{1}{n} XX^T \right) \boldsymbol{c_2} + \theta(\boldsymbol{c_2}^T \boldsymbol{c_2} - 1) + \beta \boldsymbol{c_2}^T \boldsymbol{u_1}$$

$$\boldsymbol{y_i} = L^{-1/2} U^T (\boldsymbol{x_i} - \bar{\boldsymbol{x}})$$

$$J(\mathcal{C}) = \sum_{i=1}^k \sum_{\boldsymbol{x_j} \in \mathcal{C}_i} \|\boldsymbol{x_j} - \boldsymbol{\mu_k}\|^2$$

$$d(\boldsymbol{x}, \boldsymbol{y}) = \sqrt{\sum_{i=1} (x_i - y_i)^2}$$

$$d(\boldsymbol{x}, \boldsymbol{y}) = \frac{\sum_{i=1} (x_i - \mu_x)(y_i - \mu_y)}{\sigma_x \sigma_y}$$

$$\boldsymbol{\mu_i} = \boldsymbol{\mu}(\mathcal{C}_i) = \underset{\boldsymbol{\mu} \in \mathcal{X}}{\text{argmin}} \sum_{\boldsymbol{x} \in \mathcal{C}_i} d(\boldsymbol{x}, \boldsymbol{\mu})$$

$$\boldsymbol{\mu_j} = \frac{\sum_{i=1}^n \mathbb{1}\{\boldsymbol{c_i} = j\} \boldsymbol{x_i}}{\sum_{i=1}^n \mathbb{1}\{\boldsymbol{c_i} = j\}}$$

$$J(\mathcal{C}_1, \ldots, \mathcal{C}_k) = \sum_{i=1}^k \sum_{\boldsymbol{x} \in \mathcal{C}_i} d(\boldsymbol{x}, \boldsymbol{\mu_i}) = \sum_{i=1}^n \|\boldsymbol{x_i} - \boldsymbol{\mu_{c_i}}\|^2$$